

Lesson 5: switch case

Switch case statements are a substitute for long if statements that compare to an integral value. The basic format for using switch case is outlined below.

```
switch ( value ) {  
case this:  
    Code to execute if value == this  
    break;  
case that:  
    Code to execute if value == that  
    break;  
...  
default:  
    Code to execute if value != this or that  
    break;  
}
```

The condition of a switch statement is a value. The case says that if it has the value of whatever is after that case then do whatever follows the colon. The break is used to break out of the case statements. Break is a keyword that breaks out of the code block, usually surrounded by braces, which it is in. In this case, break prevents the program from falling through and executing the code in all the other case statements. An important thing to note about the switch statement is that the case values may only be constant integral expressions. Sadly, it isn't legal to use case like this:

```
int a = 10;  
int b = 10;  
int c = 20;  
  
switch ( a ) {  
case b:  
    // Code  
    break;  
case c:  
    // Code  
    break;  
default:  
    // Code  
    break;  
}
```

The default case is optional, but it is wise to include it as it handles any unexpected cases. Switch statements serves as a simple way to write long if statements when the requirements are met. Often it can be used to process input from a user.

Below is a sample program, in which not all of the proper functions are actually declared, but which shows how one would use switch in a program.

```
#include <iostream>  
  
using namespace std;
```

```
void playgame();
void loadgame();
void playmultiplayer();

int main()
{
    int input;

    cout<<"1. Play game\n";
    cout<<"2. Load game\n";
    cout<<"3. Play multiplayer\n";
    cout<<"4. Exit\n";
    cout<<"Selection: ";
    cin>> input;
    switch ( input ) {
        case 1:                // Note the colon, not a semicolon
            playgame();
            break;
        case 2:                // Note the colon, not a semicolon
            loadgame();
            break;
        case 3:                // Note the colon, not a semicolon
            playmultiplayer();
            break;
        case 4:                // Note the colon, not a semicolon
            cout<<"Thank you for playing!\n";
            break;
        default:               // Note the colon, not a semicolon
            cout<<"Error, bad input, quitting\n";
            break;
    }
    cin.get();
}
```

This program will compile, but cannot be run until the undefined functions are given bodies, but it serves as a model (albeit simple) for processing input. If you do not understand this then try mentally putting in if statements for the case statements. Default simply skips out of the switch case construction and allows the program to terminate naturally. If you do not like that, then you can make a loop around the whole thing to have it wait for valid input. You could easily make a few small functions if you wish to test the code.